Exp1:

To implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Data set 1:

Sky	AirTemp	Humidity	Wind	Water	Forecast	PlayTennis
Sunny	Warm	Normal	Strong	Warm	Same	Yes
Sunny	Warm	High	Strong	Warm	Same	Yes
Rainy	Cold	High	Strong	Warm	Change	No
Sunny	Warm	High	Strong	Cool	Same	Yes
Rainy	Warm	Normal	Weak	Warm	Same	No
Sunny	Warm	Normal	Weak	Warm	Same	Yes

Code:

```
import pandas as pd

# Training Dataset

data = [

    ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same', 'Yes'],

    ['Sunny', 'Warm', 'High',   'Strong', 'Warm', 'Same', 'Yes'],

    ['Rainy', 'Cold', 'High',   'Strong', 'Warm', 'Change', 'No'],

    ['Sunny', 'Warm', 'High',   'Strong', 'Cool', 'Change', 'Yes'],

    ['Sunny', 'Warm', 'Normal', 'Weak',   'Warm', 'Same', 'No'],

    ['Sunny', 'Warm', 'Normal', 'Weak',   'Warm', 'Same', 'Yes']

]


# Column Names
```

```
columns = ['Sky', 'AirTemp', 'Humidity', 'Wind', 'Water', 'Forecast', 'PlayTennis']
# Create DataFrame
df = pd.DataFrame(data, columns=columns)
print("Training Dataset:\n")
print(df)
# Initialize hypothesis with the first positive example
hypothesis = None
for index, row in df.iterrows():
    if row['PlayTennis'] == 'Yes':
        hypothesis = row[:-1].tolist()
        break
# Apply FIND-S Algorithm
for index, row in df.iterrows():
    if row['PlayTennis'] == 'Yes':
        attributes = row[:-1].tolist()

        for i in range(len(hypothesis)):
            if hypothesis[i] != attributes[i]:
                hypothesis[i] = '?'

print("\nFinal Hypothesis:")
print(hypothesis)
```

output:

```
•  Training Dataset:

       Sky AirTemp Humidity    Wind Water Forecast PlayTennis
   0  Sunny    Warm   Normal  Strong  Warm     Same        Yes
   1  Sunny    Warm     High  Strong  Warm     Same        Yes
   2  Rainy    Cold     High  Strong  Warm   Change         No
   3  Sunny    Warm     High  Strong  Cool   Change        Yes
   4  Sunny    Warm   Normal    Weak  Warm     Same         No
   5  Sunny    Warm   Normal    Weak  Warm     Same        Yes

   Final Hypothesis:
   ['Sunny', 'Warm', '?', '?', '?', '?']
```

Data set 2:

Income	**CreditScore**	**Employment**	**Property**	**Age**	**LoanApproved**
High	Good	Permanent	Yes	Young	Yes
High	Good	Permanent	No	Middle	Yes
Low	Poor	Temporary	No	Young	No
Medium	Good	Permanent	Yes	Middle	Yes
High	Average	Temporary	Yes	Old	No
High	Good	Permanent	Yes	Middle	Yes

Code:

```
import pandas as pd


# Loan Approval Dataset
data = [
    ['High',   'High',    'Permanent', 'Good',    'Yes'],
    ['Low',    'Good',    'Temporary', 'Average', 'No'],
```

```
    ['High',   'High',    'Permanent', 'Good',    'Yes'],
    ['Medium', 'Average', 'Permanent', 'Poor',    'No'],
    ['High',   'Good',    'Temporary', 'Good',    'Yes']
]
# Column Names
columns = ['Income', 'CreditScore', 'Employment', 'Property',
'LoanApproved']
# Create DataFrame
df = pd.DataFrame(data, columns=columns)
print("Training Dataset:\n")
print(df)
# Initialize hypothesis with the first positive example
hypothesis = None
for index, row in df.iterrows():
    if row['LoanApproved'] == 'Yes':
        hypothesis = row[:-1].tolist()
        break
# Apply FIND-S Algorithm
for index, row in df.iterrows():
    if row['LoanApproved'] == 'Yes':
        attributes = row[:-1].tolist()

        for i in range(len(hypothesis)):
            if hypothesis[i] != attributes[i]:
                hypothesis[i] = '?'
```

```
# Display Final Hypothesis
print("\nFinal Hypothesis:")
print(hypothesis)
```

Output:

```
Training Dataset:

   Income CreditScore Employment Property LoanApproved
0    High        High  Permanent     Good          Yes
1     Low        Good  Temporary  Average           No
2    High        High  Permanent     Good          Yes
3  Medium     Average  Permanent     Poor           No
4    High        Good  Temporary     Good          Yes

Final Hypothesis:
['High', '?', '?', 'Good']
```

Data set 3:

CGPA	Communication	Internship	Programming	Aptitude	Placed
High	Good	Yes	Good	High	Yes
High	Excellent	Yes	Good	High	Yes
Medium	Average	No	Average	Medium	No
High	Good	Yes	Excellent	High	Yes
Low	Poor	No	Average	Low	No
High	Good	Yes	Good	Medium	Yes

Code:

```
import pandas as pd

# Student Placement Dataset
data = [
    ['High',   'Good',      'Yes', 'Good',      'High',   'Yes'],
    ['High',   'Excellent', 'Yes', 'Good',      'High',   'Yes'],
    ['Medium', 'Average',   'No',  'Average',   'Medium', 'No'],
    ['High',   'Good',      'Yes', 'Excellent', 'High',   'Yes'],
    ['Low',    'Poor',      'No',  'Average',   'Low',    'No'],
    ['High',   'Good',      'Yes', 'Good',      'Medium', 'Yes']
]

# Column Names
columns = [
    'CGPA',
    'Communication',
    'Internship',
    'Programming',
    'Aptitude',
    'Placed'
]

# Create DataFrame
df = pd.DataFrame(data, columns=columns)
```

```python
print("Training Dataset:\n")
print(df)

# Initialize hypothesis with the first positive example
hypothesis = None

for index, row in df.iterrows():
    if row['Placed'] == 'Yes':
        hypothesis = row[:-1].tolist()
        break

# FIND-S Algorithm
for index, row in df.iterrows():
    if row['Placed'] == 'Yes':
        attributes = row[:-1].tolist()

        for i in range(len(hypothesis)):
            if hypothesis[i] != attributes[i]:
                hypothesis[i] = '?'

# Display Final Hypothesis
print("\nFinal Hypothesis:")
print(hypothesis)
```

Output:

```
•••  Training Dataset:

       CGPA Communication Internship Programming Aptitude Placed
    0    High          Good        Yes        Good     High    Yes
    1    High     Excellent        Yes        Good     High    Yes
    2  Medium       Average         No     Average   Medium     No
    3    High          Good        Yes   Excellent     High    Yes
    4     Low          Poor         No     Average      Low     No
    5    High          Good        Yes        Good   Medium    Yes

    Final Hypothesis:
    ['High', '?', 'Yes', '?', '?']
```

Data set 4:

Fever	Cough	Headache	BodyPain	Fatigue	Disease
Yes	Yes	Yes	Yes	Yes	Positive
Yes	Yes	No	Yes	Yes	Positive
No	Yes	Yes	No	No	Negative
Yes	Yes	Yes	No	Yes	Positive
No	No	Yes	Yes	No	Negative
Yes	Yes	No	No	Yes	Positive

Code:

```
import pandas as pd


# Disease Diagnosis Dataset
data = [
    ['Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'Positive'],
    ['Yes', 'Yes', 'No',  'Yes', 'Yes', 'Positive'],
```

```
    ['No',  'Yes', 'Yes', 'No',  'No',  'Negative'],
    ['Yes', 'Yes', 'Yes', 'No',  'Yes', 'Positive'],
    ['No',  'No',  'Yes', 'Yes', 'No',  'Negative'],
    ['Yes', 'Yes', 'No',  'No',  'Yes', 'Positive']
]

# Column Names
columns = [
    'Fever',
    'Cough',
    'Headache',
    'BodyPain',
    'Fatigue',
    'Disease'
]

# Create DataFrame
df = pd.DataFrame(data, columns=columns)

print("Training Dataset:\n")
print(df)

# Initialize hypothesis with the first positive example
hypothesis = None
```

```
for index, row in df.iterrows():
    if row['Disease'] == 'Positive':
        hypothesis = row[:-1].tolist()
        break

# FIND-S Algorithm
for index, row in df.iterrows():
    if row['Disease'] == 'Positive':
        attributes = row[:-1].tolist()

        for i in range(len(hypothesis)):
            if hypothesis[i] != attributes[i]:
                hypothesis[i] = '?'

# Display Final Hypothesis
print("\nFinal Hypothesis:")
print(hypothesis)
```

Output:

```
Training Dataset:

  Fever Cough Headache BodyPain Fatigue   Disease
0   Yes   Yes      Yes      Yes     Yes  Positive
1   Yes   Yes       No      Yes     Yes  Positive
2    No   Yes      Yes       No      No  Negative
3   Yes   Yes      Yes       No     Yes  Positive
4    No    No      Yes      Yes      No  Negative
5   Yes   Yes       No       No     Yes  Positive

Final Hypothesis:
['Yes', 'Yes', '?', '?', 'Yes']
```

Data set 5:

ContainsLink	ContainsOffer	UnknownSender	Attachment	ManyRecipients	Spam
Yes	Yes	Yes	No	Yes	Yes
Yes	Yes	Yes	Yes	Yes	Yes
No	No	No	No	No	No
Yes	No	Yes	No	Yes	Yes
No	Yes	No	Yes	No	No
Yes	Yes	Yes	No	No	Yes

Code:

```
import pandas as pd


# Email Spam Detection Dataset
data = [
    ['Yes', 'Yes', 'Yes', 'No',  'Yes', 'Yes'],
    ['Yes', 'Yes', 'Yes', 'Yes', 'Yes', 'Yes'],
    ['No',  'No',  'No',  'No',  'No',  'No'],
```

```
    ['Yes', 'No',  'Yes', 'No',  'Yes', 'Yes'],
    ['No',  'Yes', 'No',  'No',  'Yes', 'No'],
    ['Yes', 'Yes', 'Yes', 'No',  'No',  'Yes']
]

# Column Names
columns = [
    'ContainsLink',
    'ContainsOffer',
    'UnknownSender',
    'Attachment',
    'ManyRecipients',
    'Spam'
]

# Create DataFrame
df = pd.DataFrame(data, columns=columns)

print("Training Dataset:\n")
print(df)

# Initialize hypothesis with the first positive example
hypothesis = None
```

```
for index, row in df.iterrows():
    if row['Spam'] == 'Yes':
        hypothesis = row[:-1].tolist()
        break

# FIND-S Algorithm
for index, row in df.iterrows():
    if row['Spam'] == 'Yes':
        attributes = row[:-1].tolist()

        for i in range(len(hypothesis)):
            if hypothesis[i] != attributes[i]:
                hypothesis[i] = '?'

# Display Final Hypothesis
print("\nFinal Hypothesis:")
print(hypothesis)
```

Output:

```
Training Dataset:

  ContainsLink ContainsOffer UnknownSender Attachment ManyRecipients Spam
0          Yes           Yes           Yes         No            Yes  Yes
1          Yes           Yes           Yes        Yes            Yes  Yes
2           No            No            No         No             No   No
3          Yes            No           Yes         No            Yes  Yes
4           No           Yes            No         No            Yes   No
5          Yes           Yes           Yes         No             No  Yes

Final Hypothesis:
['Yes', '?', 'Yes', '?', '?']
```